

Liu Peng

Gender: Male Objective: Backend Engineer Location: Suzhou

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Education

2013.09 - 2017.06 Sun Yat-sen University Software Engineering (B.S.)

Courses: Computer Networks, Computer Organization, Machine Learning, Compilers, Database Systems, Software Engineering

GPA 3.5/4.0, top 20% in major; merit scholarship

Summary

Backend engineer with several years of Java experience and an ~1-year career gap, self-studied and contributed to open source during the break.

Work Experience

2021.04 - Present Zhihu Backend Engineer

- Tuned Redis caching for hot keys and cache breakdown with local cache and distributed locks, raising hit rate above 90%, reused across multiple business lines.

- Fixed a log-collection failure caused by agent backpressure, restoring stability after tuning batch and buffer settings.

- Owned backend of core transaction and order services, splitting the monolith into order, settlement and risk-control microservices, improving end-to-end delivery quality.

Career gap: 2020.05 - 2021.04 (break for personal reasons; self-studied and contributed to open source)

2017.07 - 2020.05 Lalamove Backend Engineer

- Optimized a file-sort slow SQL on a hot API, added a composite index via gh-ost online DDL, reducing P99 from 1500ms to 420ms.

- Investigated rising instance memory rss using GC logs, heap dump and arthas, found an unbounded cache and cut memory by 20%, and related standards were adopted by the team.

- Led a production incident review with root-cause analysis and drove an alert-tiering and on-call rotation mechanism, significantly lowering maintenance cost.

Projects

Distributed Payment & Settlement Platform (Spring Boot + MySQL + Kafka + Redis)

- Recorded an audit trace id for every payment state transition for compliance, and it became a reusable team best practice.

- Implemented daily reconciliation across payment gateway, ledger and settlement files with auto-correction.

- Applied idempotency and eventual consistency to avoid duplicate charges on retry.

- Used sharding and read-write splitting to keep single-table rows within tens of millions, and key metrics improved steadily.

Enterprise Auth & Gateway Platform (Spring Cloud Gateway + OAuth2)

- Shipped a unified SDK to lower integration cost across teams.

- Integrated OAuth2 and RBAC with multi-tenant isolation.
 - Unified auth, rate limiting, gray routing and tracing as reusable platform capabilities, recognized by stakeholders and peers.
 - Added circuit breaking to isolate faulty downstreams, and related standards were adopted by the team.
- High-Concurrency Flash-Sale & Inventory System (Java + Redis + RocketMQ)
- Built circuit breaking and fallback for downstream failures.
 - Pre-deducted inventory in Redis with async DB writes to absorb traffic spikes, reused across multiple business lines.
 - Stayed stable at 5000 QPS in load tests with error rate below 0.1%, and it became a reusable team best practice.
 - Designed distributed locks and token-bucket rate limiting to prevent oversell.

Highlights

- Drove a focused effort on Redis caching and distributed locks, reducing related issues by about 24%
- Continuously optimized cross-team communication, cutting key-path latency from 1500ms to 420ms
- Led a major technical effort involving MySQL indexing and slow-query tuning, staying stable at 5000 QPS peak
- Owned core modules that steadily served 4260K daily visits with 99.9%+ availability
- Established metrics and reviews around requirement analysis and documentation to keep improving delivery quality

Skills

Core skills: Java, Redis caching and distributed locks, MySQL indexing and slow-query tuning

Proficient in: MySQL, Docker and containerization, Backend

Familiar with: Distributed transactions and consistency, Kafka/RocketMQ messaging, JVM tuning and troubleshooting, Spring

Self Evaluation

- Curious about new tech, continuously learning LLM and cloud-native topics and applying them in practice
- Business-minded, good at balancing technical design against delivery timelines